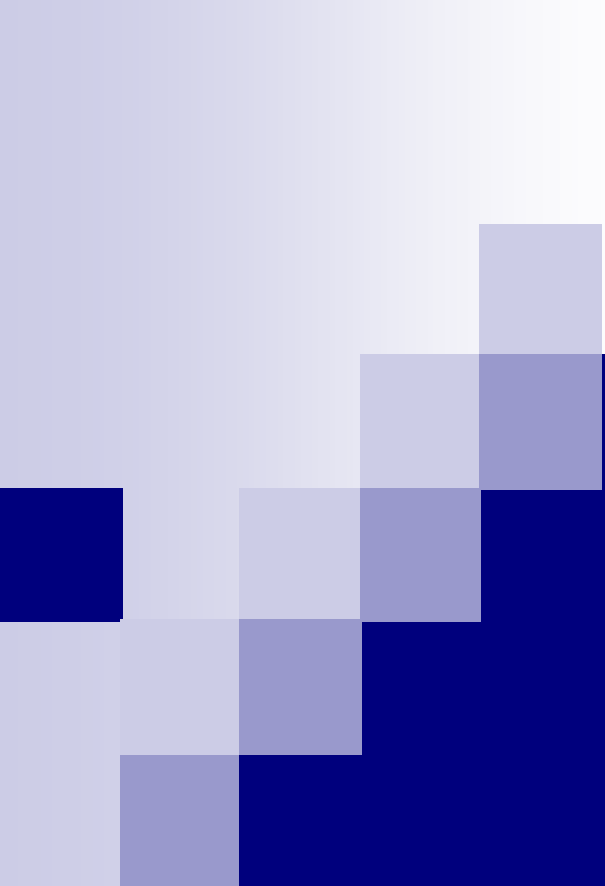
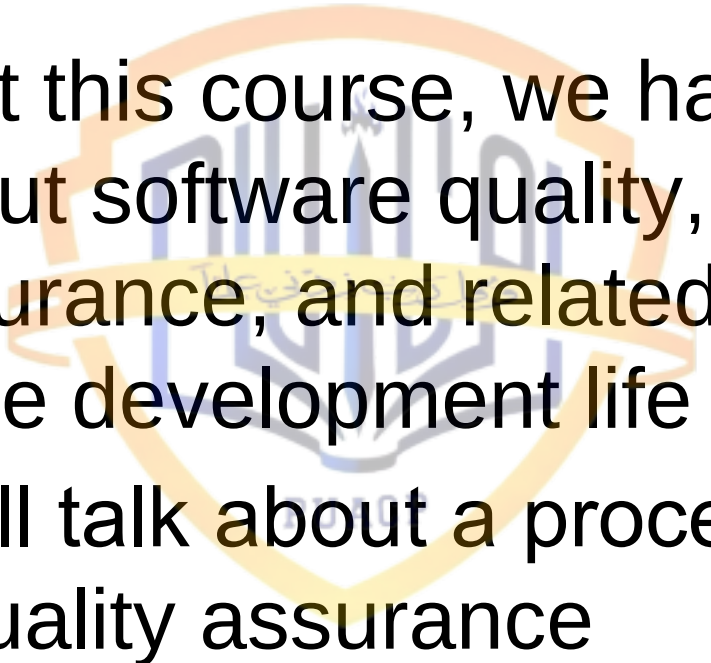
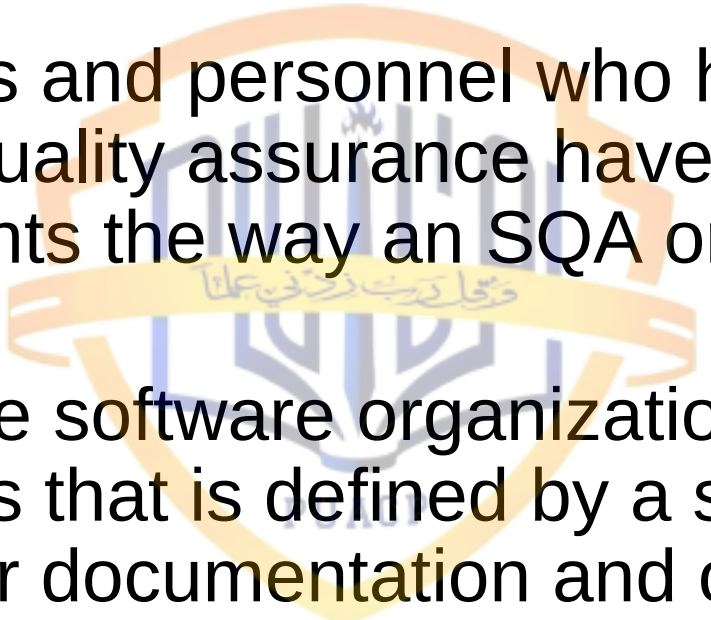




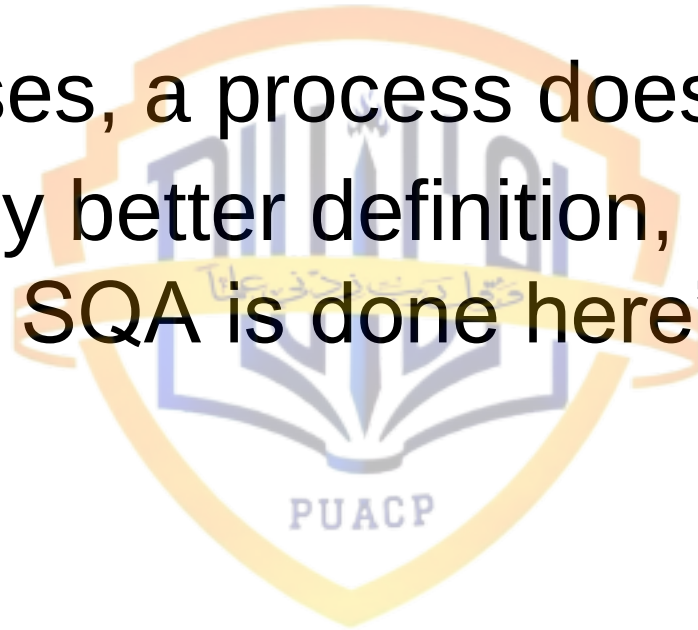
A Process Model for Software Quality Assurance

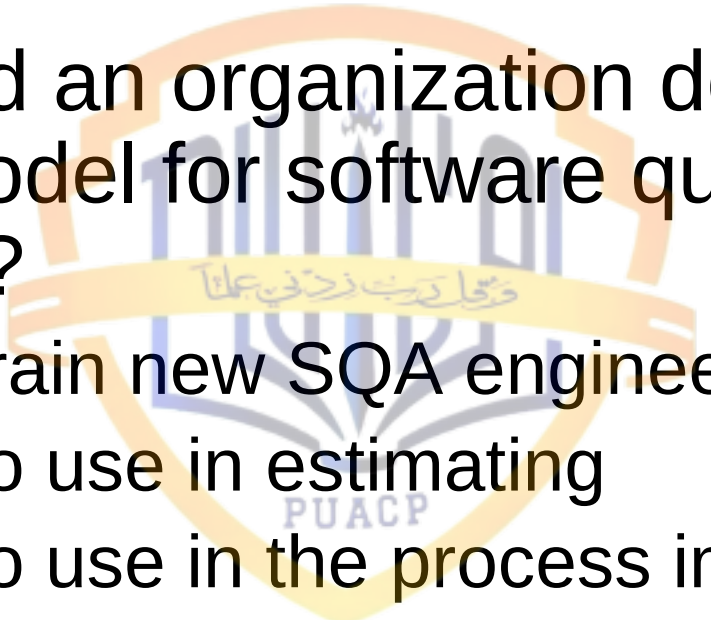
Lecture # 44

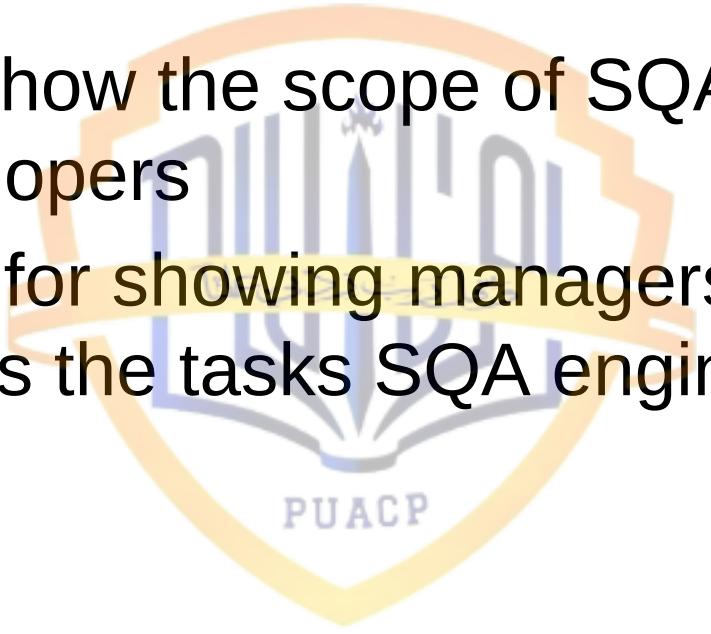
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- Throughout this course, we have been talking about software quality, software quality assurance, and related activities in the software development life cycle
 - Today, we'll talk about a process model of software quality assurance

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- All managers and personnel who have worked in a software quality assurance have a process that represents the way an SQA organization operates
 - Some mature software organizations have an SQA process that is defined by a set of standards for documentation and operations. Other smaller firms may have a less well-defined and more ad-hoc process; the primary repository for information regarding this process is in the minds of its SQA personnel

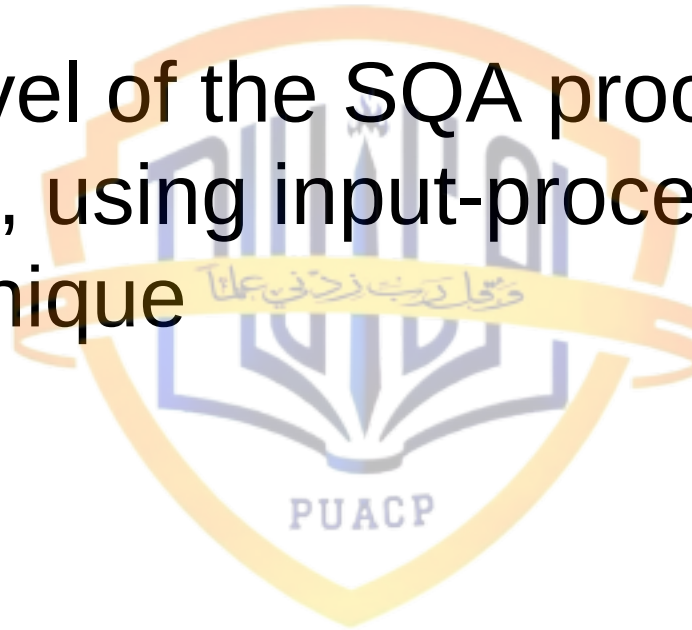
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- In both cases, a process does exist
 - Lacking any better definition, the process is “the way SQA is done here”

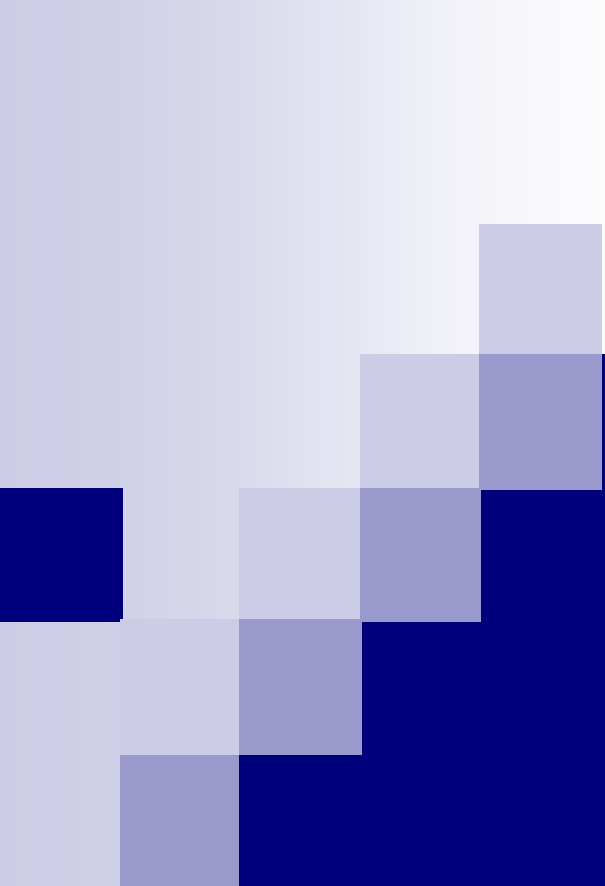


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- What would an organization do with a process model for software quality assurance?
 - A tool to train new SQA engineers
 - A model to use in estimating
 - A model to use in the process improvement
 - A model to use to select or develop a set of standards and templates for SQA documentation

- 
- A tool to show the scope of SQA activities to new developers
 - A graphic for showing managers and developers the tasks SQA engineers perform
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- The top-level of the SQA process is shown graphically, using input-process-output (IPO) technique

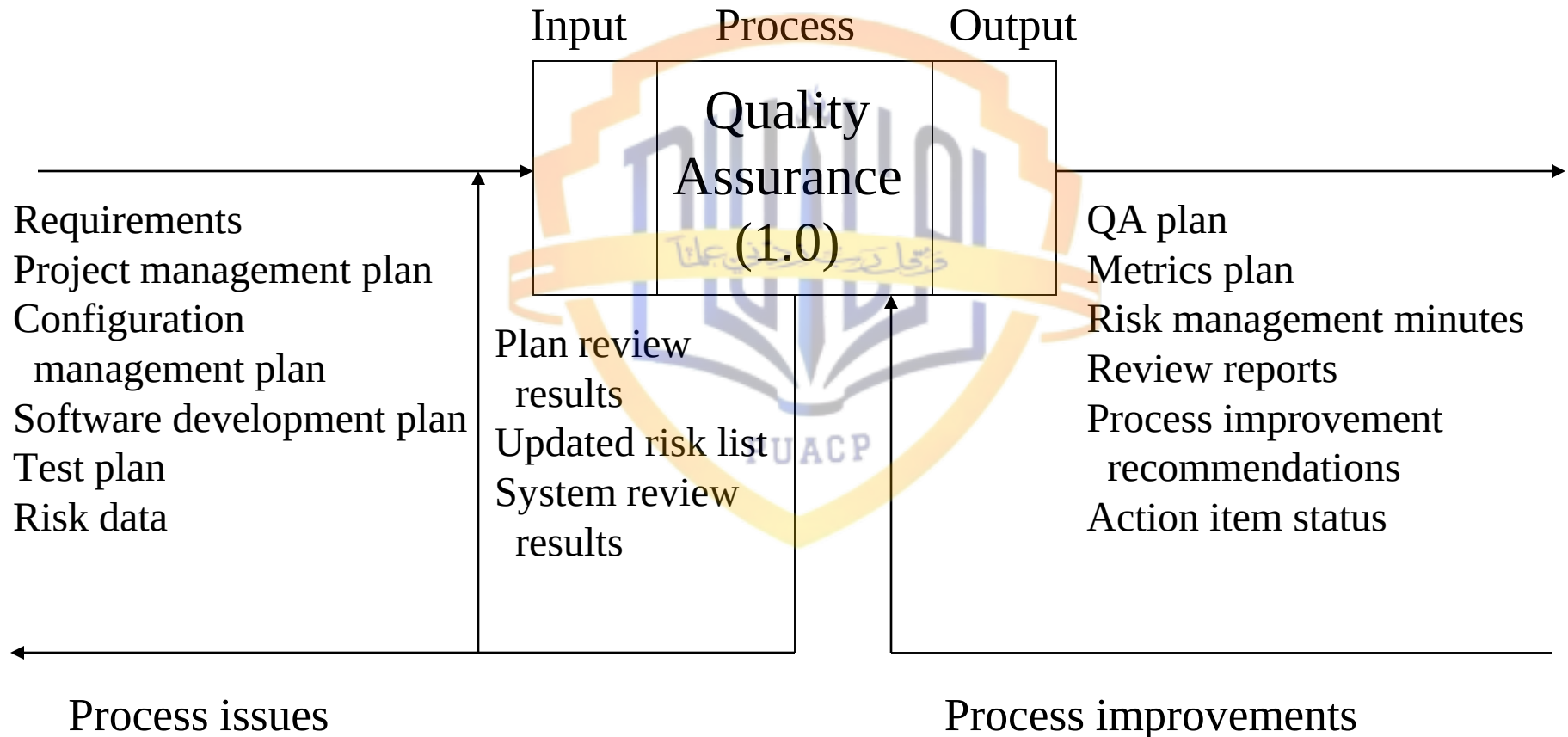




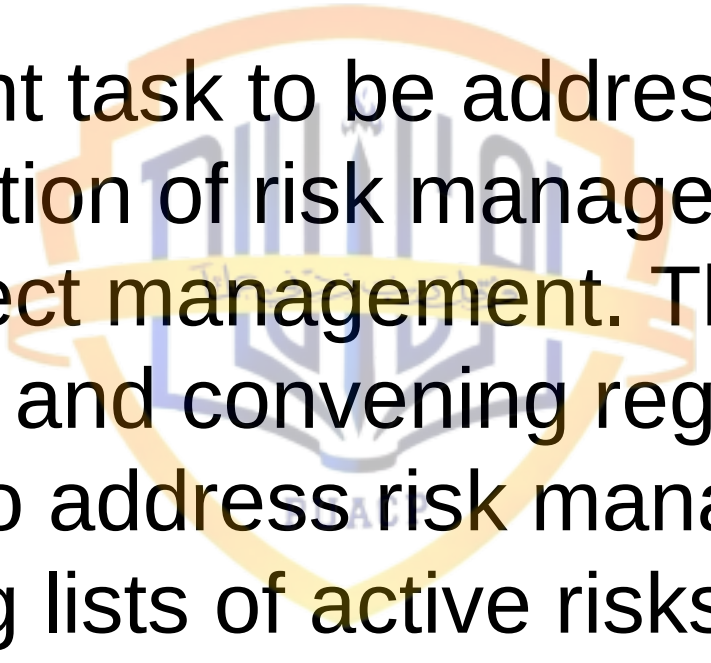
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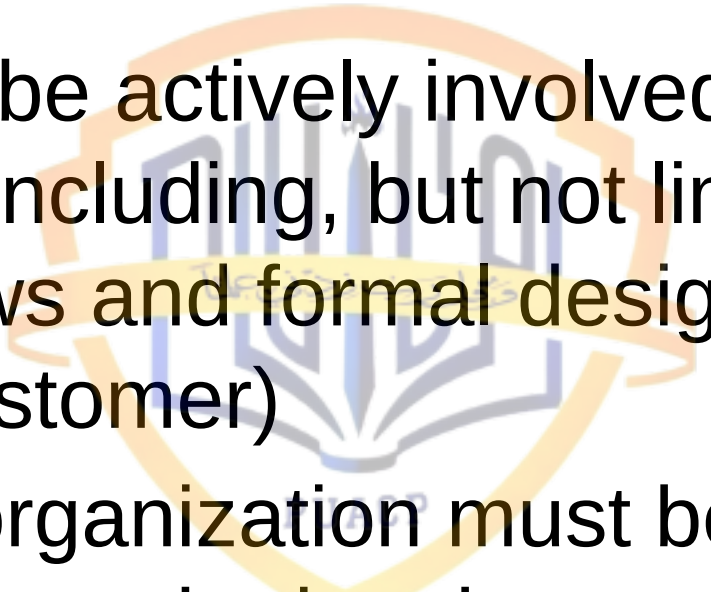
Software Quality Assurance
Level 0 IPO Diagram

Software Quality Assurance Level 0 IPO Diagram

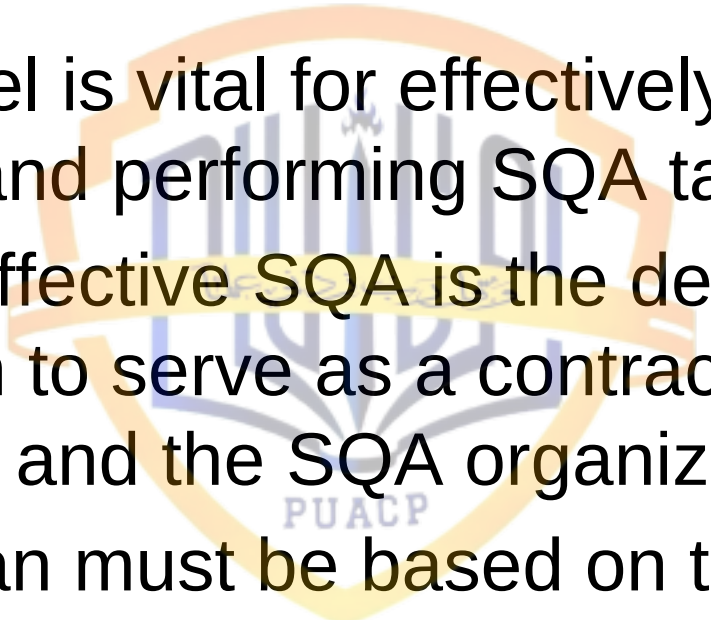


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- The SQA process is based on relationships, interfaces, and tasks that are defined in other project plans (for example, project management plan, configuration management plan)

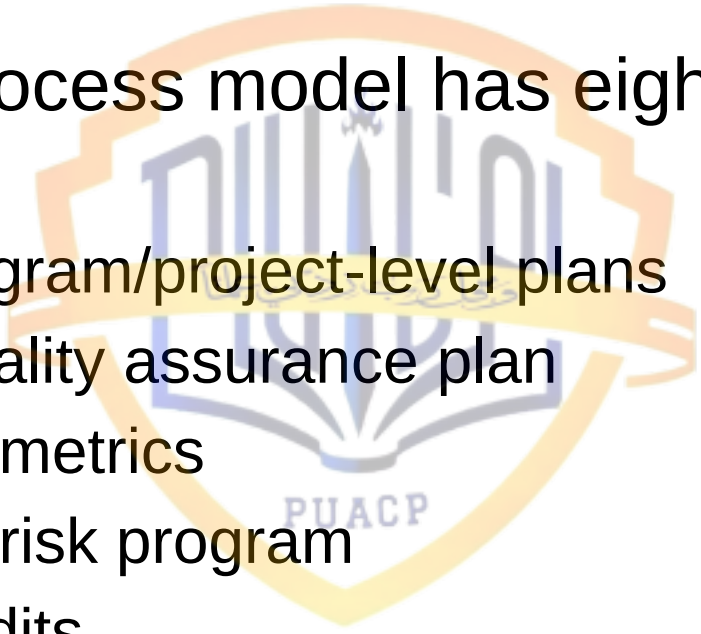
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- A significant task to be addressed by SQA is coordination of risk management to assist project management. This involves scheduling and convening regular meetings to address risk management, maintaining lists of active risks, and tracking status of risk mitigation plans for project management

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- SQA must be actively involved in review meetings (including, but not limited to, peer reviews and formal design reviews with the customer)
 - The SQA organization must be actively involved in monitoring the test program on the project

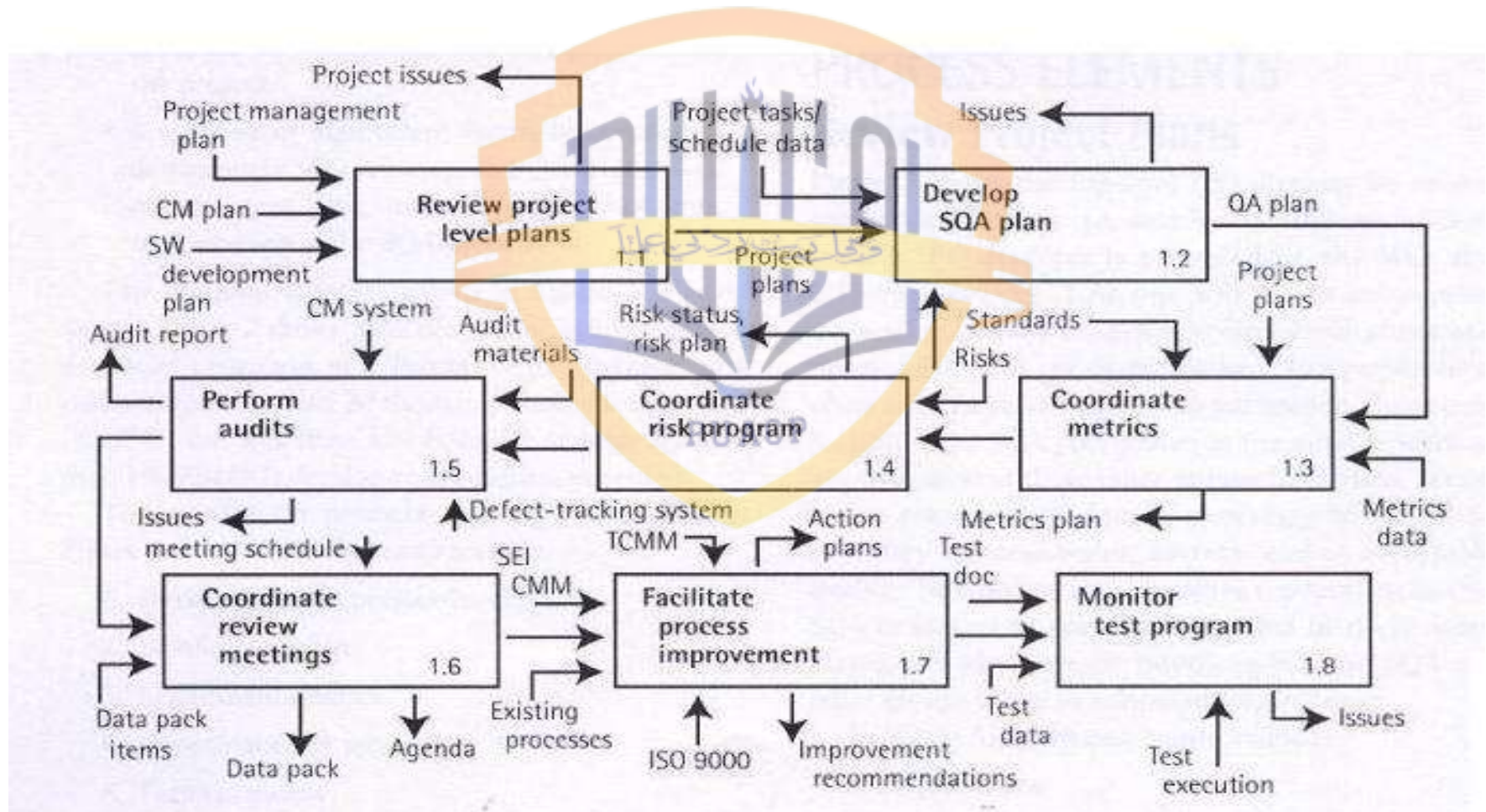
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- The number of significant formally controlled documents (for example, SQA plan, risk management plan, metrics plan) result from performance of the SQA

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- Such a model is vital for effectively identifying, estimating, and performing SQA tasks
 - The key to effective SQA is the development of an SQA plan to serve as a contract between the project team and the SQA organization
 - The SQA plan must be based on the process model that identifies the specific tasks SQA needs to perform to effectively support the project team

- This SQA process model has eight sub-processes

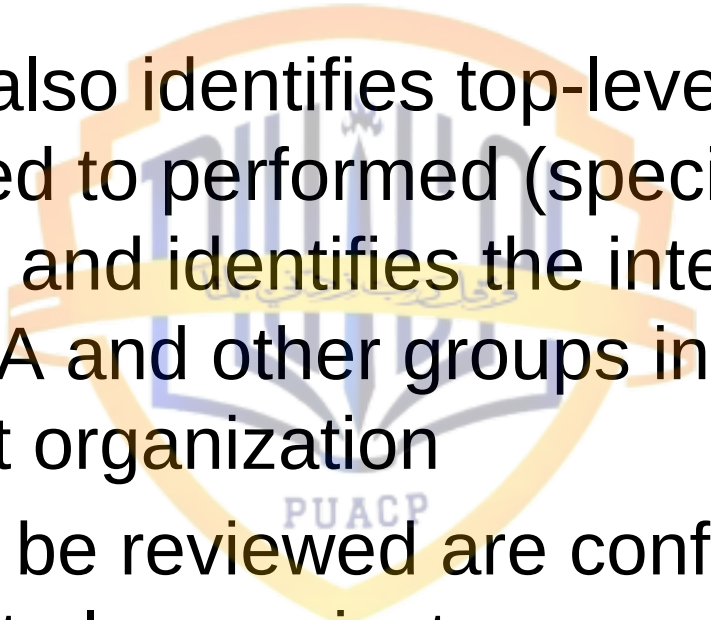
- 
- ☐ Review program/project-level plans
 - ☐ Develop quality assurance plan
 - ☐ Coordinate metrics
 - ☐ Coordinate risk program
 - ☐ Perform audits
 - ☐ Coordinate review meetings
 - ☐ Facilitate process improvement
 - ☐ Monitor test program

Software Quality Assurance Level 1 IPO Diagram

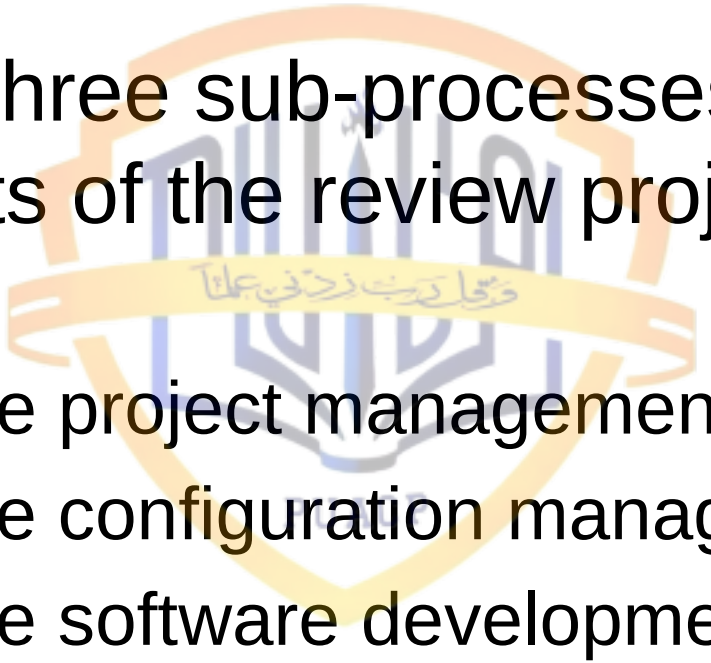


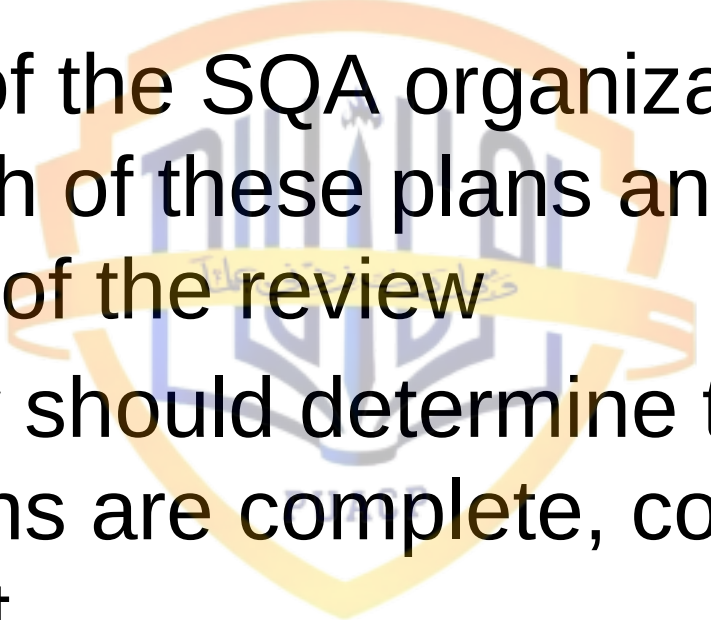
1. Review Project Plans

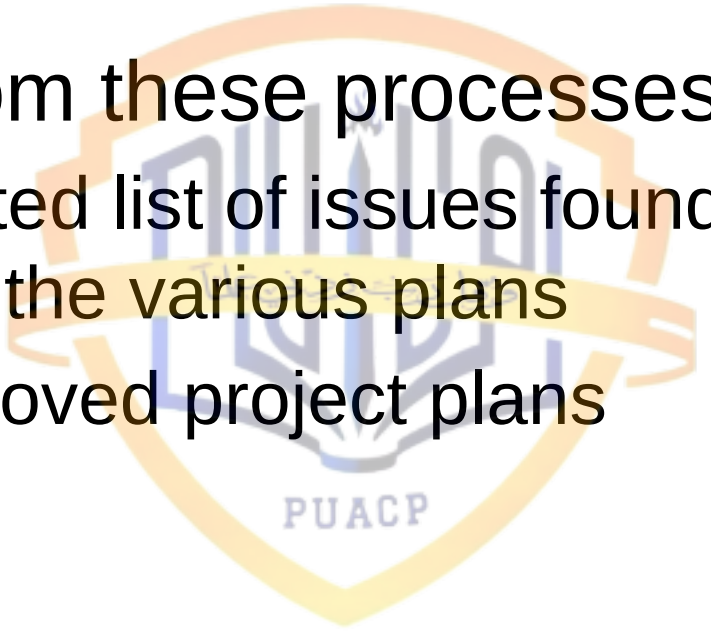
- The project-level plans must be reviewed prior to publication of SQA plan
- Since the publication of the SQA plan occurs in the same time frame as publication of these other project-level plans, review of the project-level plans is necessary to determine that they are consistent, correct, and of acceptable quality

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- This review also identifies top-level tasks that SQA is tasked to performed (specified in these other plans), and identifies the interfaces between SQA and other groups in the development organization
 - The plans to be reviewed are configuration management plan, project management plan, software development plan, configuration system, and standards/templates

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- The test plan is not reviewed at this point in the project; it is not normally written until after the requirements are generated. Thus, the test plan is developed later than the document review that is taking place in this process

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- There are three sub-processes that are components of the review project-plans process
 - Review the project management plan
 - Review the configuration management plan
 - Review the software development plan

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- Members of the SQA organization should review each of these plans and report on the results of the review
 - The review should determine that the project plans are complete, consistent, and correct

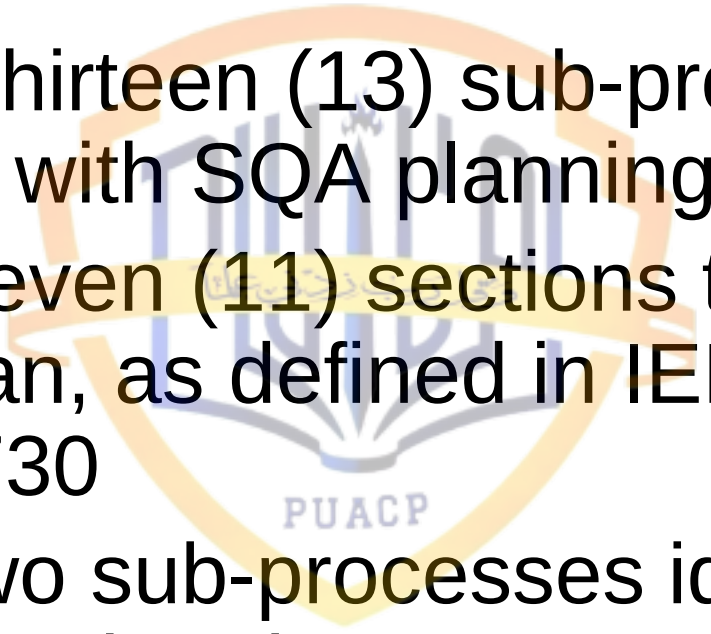
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- Outputs from these processes are simple
 - Documented list of issues found while reviewing the various plans
 - SQA-approved project plans

2. Develop QA Plan

- As a result of developing SQA plan, SQA tasks to be performed on the program or project are formally documented
- The SQA plan provides a road map to the activities that the SQA organization will perform
- Essentially, the SQA plan serves as a formal contract between the SQA organization and the other groups on the project, specifying what tasks will be performed, according to what schedule

- Inputs for this activities are

- Software development life cycle, as defined in the project management plan
- Project plans
 - Project management plan
 - CM plan
 - Software development plan
- Schedules
- Standards (specifically, IEEE-Standard-730, Standard for Software Quality Assurance Plans)

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- There are thirteen (13) sub-processes associated with SQA planning
 - The first eleven (11) sections that make up an SQA plan, as defined in IEEE-Standard-730
 - The final two sub-processes identify the tasks required to document, assemble, and review the SQA plan

Sub-Processes - 1

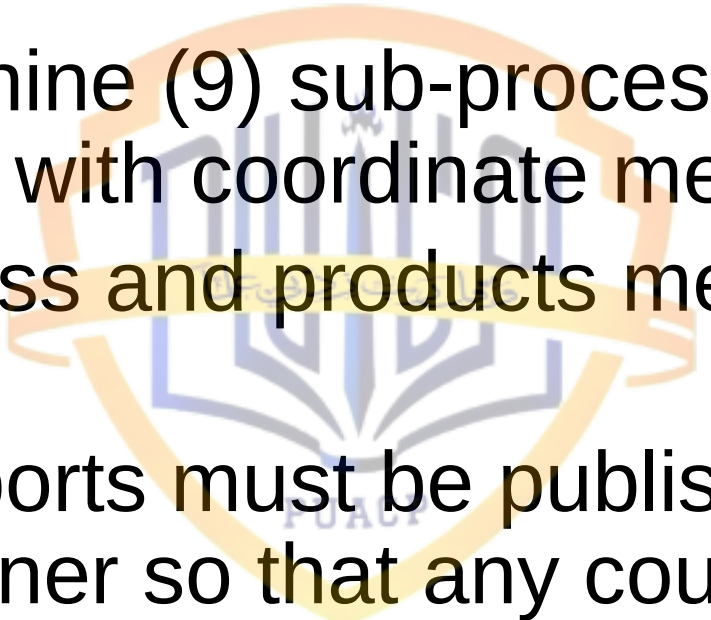
- Develop QA plan introduction
- Create management section
- Identify documentation requirements
- Identify standards
- Specify reviews and audits
- Review CM interface
- Review defect reporting

Sub-Processes - 2

- Develop metrics strategy
- Identify tools and techniques
- Define supplier control
- Define records approach
- Document SQA plan
- Review and approve SQA plan

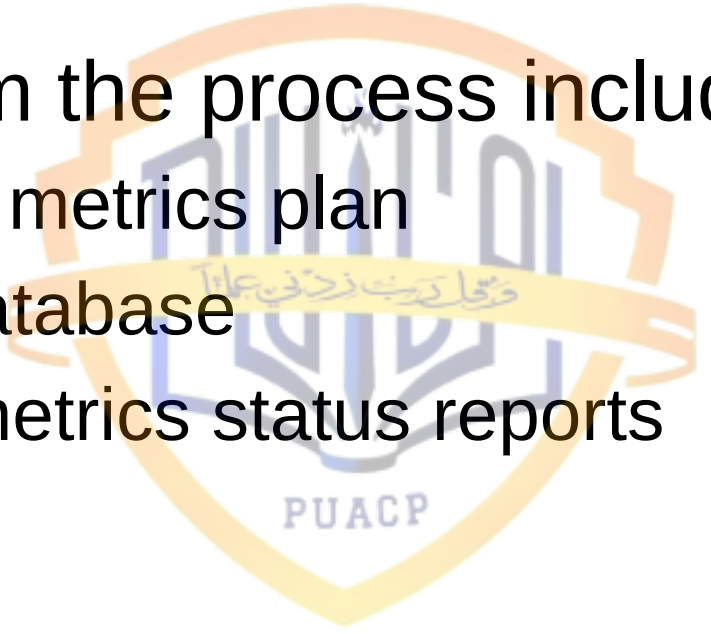
3. Coordinate Metrics

- This sub-process defines the activities and tasks that SQA performs while coordinating and collecting metrics
- Inputs are
 - Project management plan
 - Defect tracking system
 - Metrics database
 - Product metrics data
 - Process metrics data
 - Standards

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- There are nine (9) sub-processes associated with coordinate metrics
 - Both process and products metrics are addressed
 - Metrics reports must be published in a timely manner so that any course of corrections indicated by the metrics can be made quickly

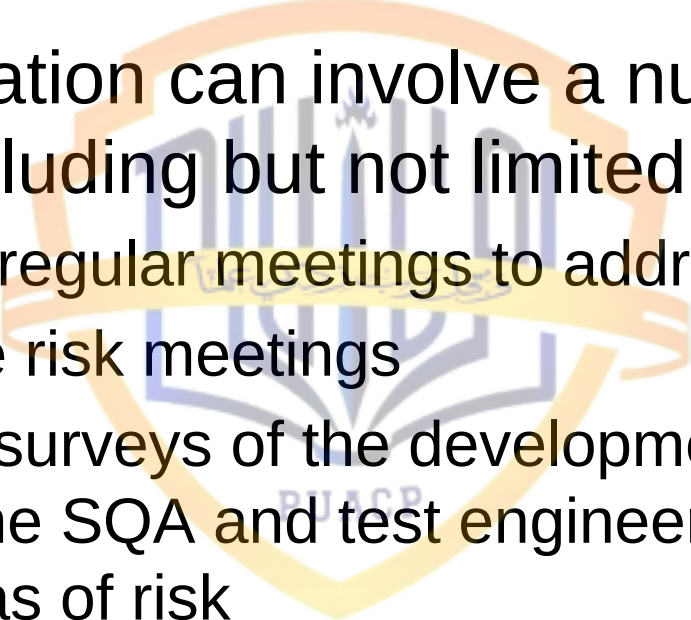
Sub-Processes

- Develop metrics strategy
- Create metrics database schema
- Document metrics plan
- Review metrics plan
- Collect measurement data
- Compute metrics
- Evaluate trends
- Issue metrics report
- Update metrics process and plans

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- Output from the process include
 - Approved metrics plan
 - Metrics database
 - Regular metrics status reports

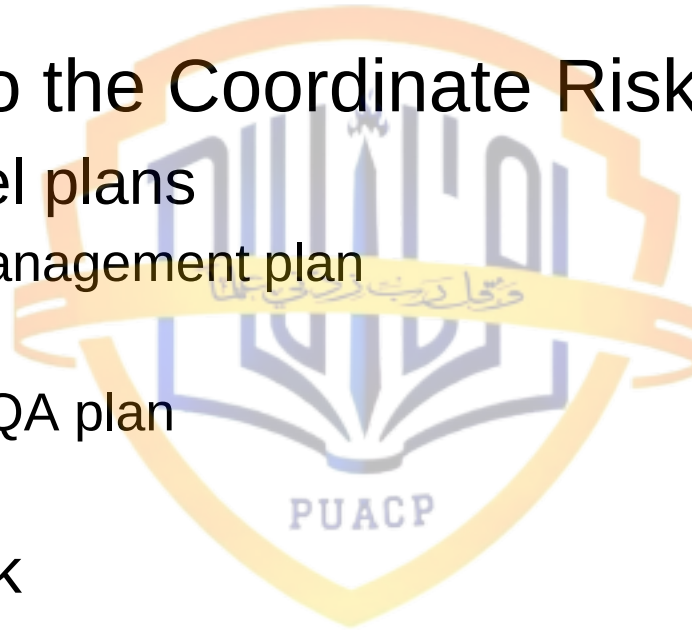
4. Coordinate Risk Program

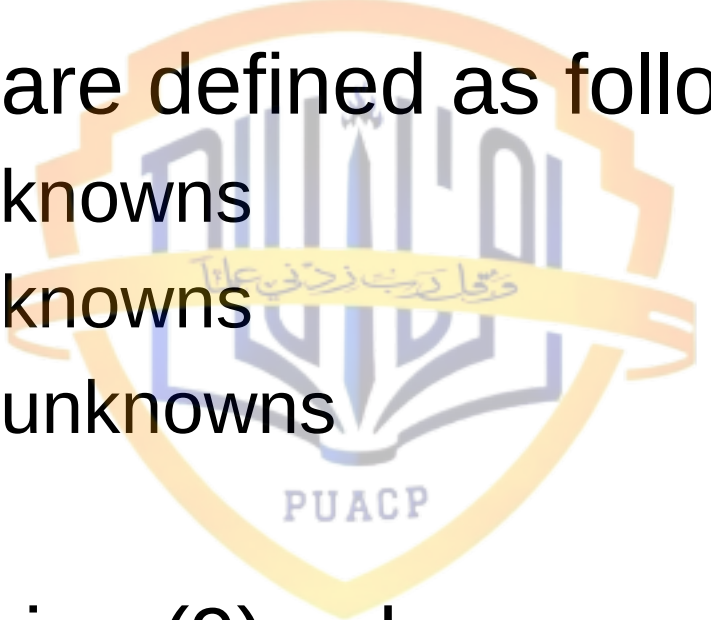
- A significant activity performed by an SQA organization is coordination of a formal risk management program on the project
- One must remember that the ownership of risk management belongs to upper-level project management
- SQA can provide invaluable assistance to management, however, through coordination and day-to-day administration of risk management

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- This coordination can involve a number of activities, including but not limited to:
 - Scheduling regular meetings to address risk issues
 - Chairing the risk meetings
 - Performing surveys of the development team (including the SQA and test engineering team) to identify areas of risk
 - Maintaining a database of risk issues
 - Collecting and tracking risk reduction plans

4. Coordinate Risk Program

- The inputs to the Coordinate Risk Program are
 - Project-level plans
 - Project management plan
 - CM plan
 - Software QA plan
 - Test plan
 - Process risk
 - Product risk
 - Risk database



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- Risk types are defined as follow
 - Known unknowns
 - Unknown knowns
 - Unknown unknowns
 - There are nine (9) sub-processes associated with Coordinate Risk Program

Sub-Processes

- Develop risk plan
- Review risk plan
- Evaluate plans, schedules for risk
- Collect process and product risks
- Establish risk database
- Perform risk assessment
- Coordinate risk control
- Coordinate risk meetings
- Issue risk reports

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- Outputs from the coordinate risk program process include
 - Risk process
 - Risk status
 - Risk management plan
 - Minutes from risk meetings
 - Risk action-item lists

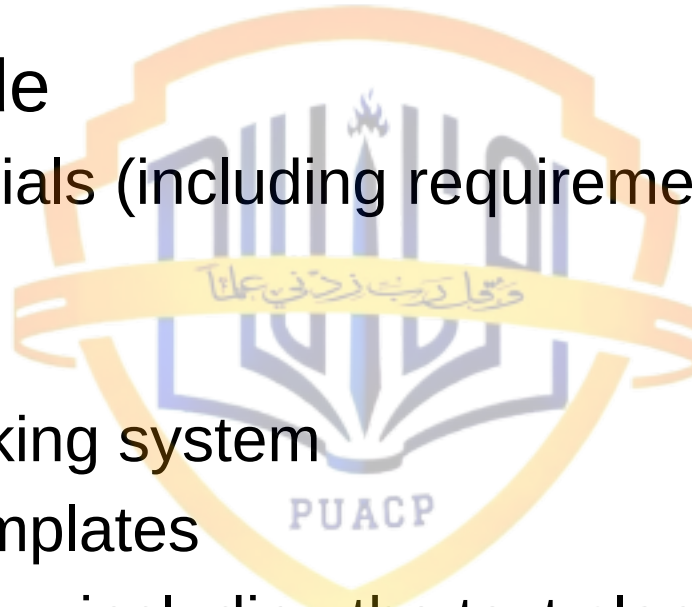
5. Perform Audits

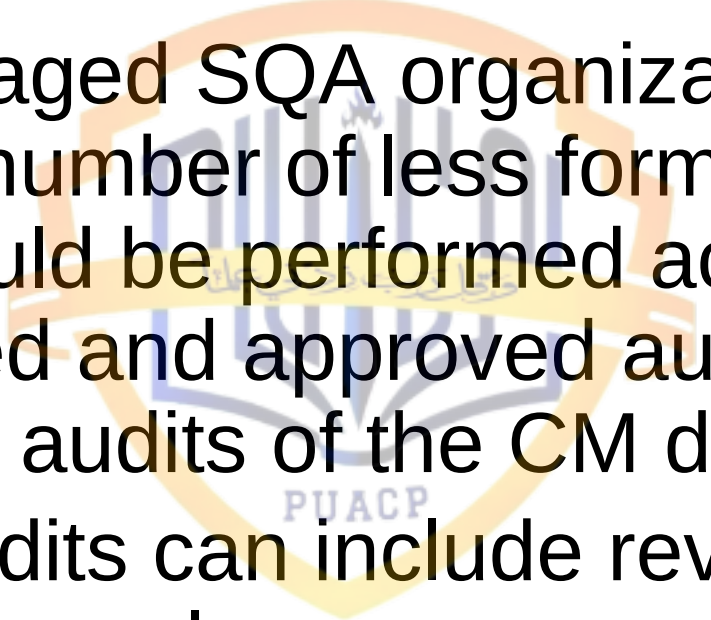
- The process for perform audits is one of the few activities remaining in modern quality assurance that can be described as a classic “policeman” activity



■ Inputs include

- ☐ Audit materials (including requirements, criteria, checklists)
- ☐ CM system
- ☐ Defect tracking system
- ☐ Process templates
- ☐ Project plans, including the test plan(s)
- ☐ Risks
- ☐ Standards and templates



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- A fully engaged SQA organization will also perform a number of less formal audits. These should be performed according to a documented and approved audit plan, and will include audits of the CM database
 - Product audits can include review of operator manuals
 - There are nine (9) sub-processes

Sub-Processes

- Review project plans
- Develop audit plan
- Review audit plan
- Establish audit database
- Perform process audits
- Perform product audits
- Perform physical configuration audit (PCA)
- Perform functional configuration audit (FCA)
- Update audit process

- Outputs from this include

- ☐ Audit checklists
- ☐ Audit issues
- ☐ Audit reports
- ☐ Action items
- ☐ Updated audit process



6. Coordinate Review Meetings

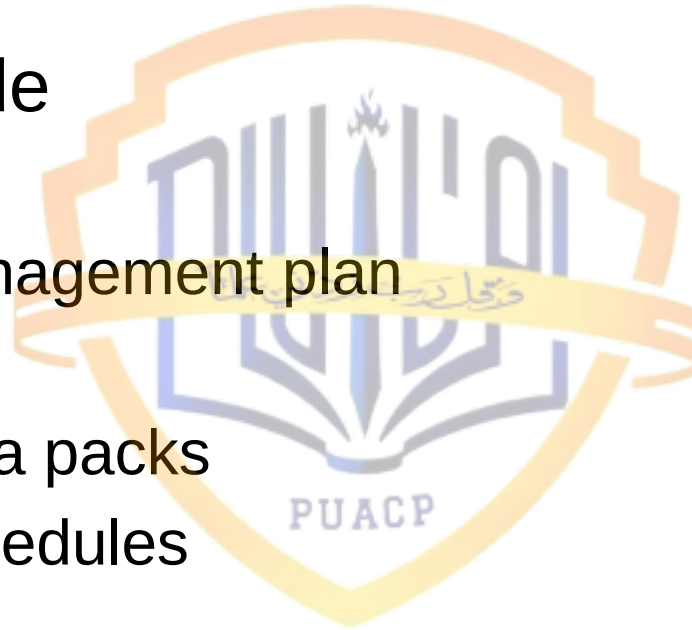
- An SQA organization is usually chartered with coordinating review meetings, on a project/program where these meetings are formally controlled
- During reviews, a specific task for SQA personnel is to evaluate the process used in performing the reviews and report on that evaluation to management



■ Inputs include

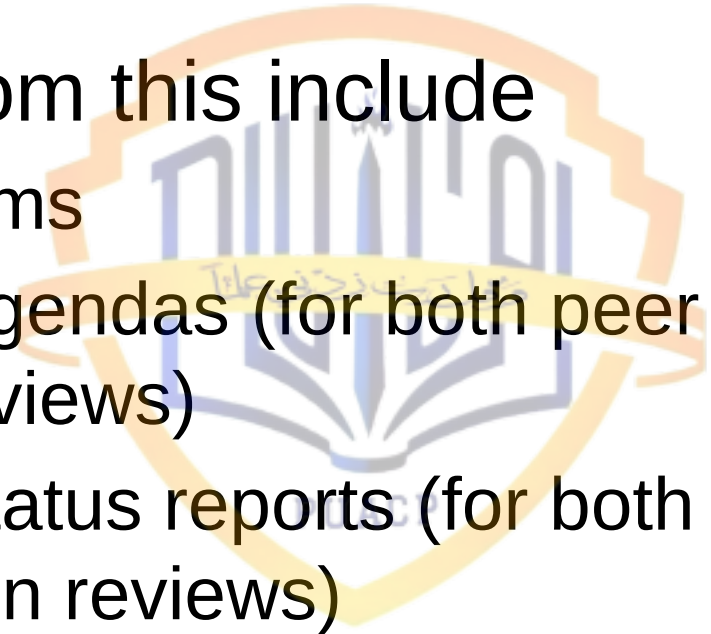
- ☐ Checklists
- ☐ Project management plan
- ☐ Test plan
- ☐ Review data packs
- ☐ Review schedules
- ☐ Standards
- ☐ Templates

■ There are nine (9) sub-processes



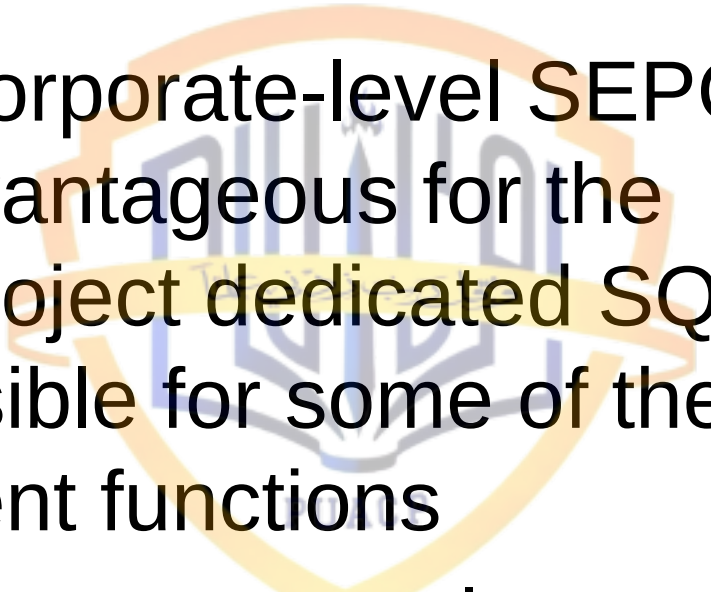
Sub-Processes

- Verify peer review schedule
- Develop (peer review) agenda template
- Support peer review meetings
- Track (peer review) action items
- Verify design review schedule
- Develop design review agenda template
- Coordinate design review data packs
- Support design review meetings
- Track design review action items

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- Outputs from this include
 - Action items
 - Review agendas (for both peer reviews and design reviews)
 - Review status reports (for both peer reviews and design reviews)

7. Facilitate Process Improvement

- A critical function of SQA organization in the current age is to facilitate process improvement
- This sub-process is only applicable for an SQA group when the program/project organization does not have, or does not have access to, an SEPG

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- Even if a corporate-level SEPG exists, it can be advantageous for the program/project dedicated SQA group to be responsible for some of the process improvement functions
 - That way, more general corporate-level initiatives can be tailored and adjusted to the needs of specific projects

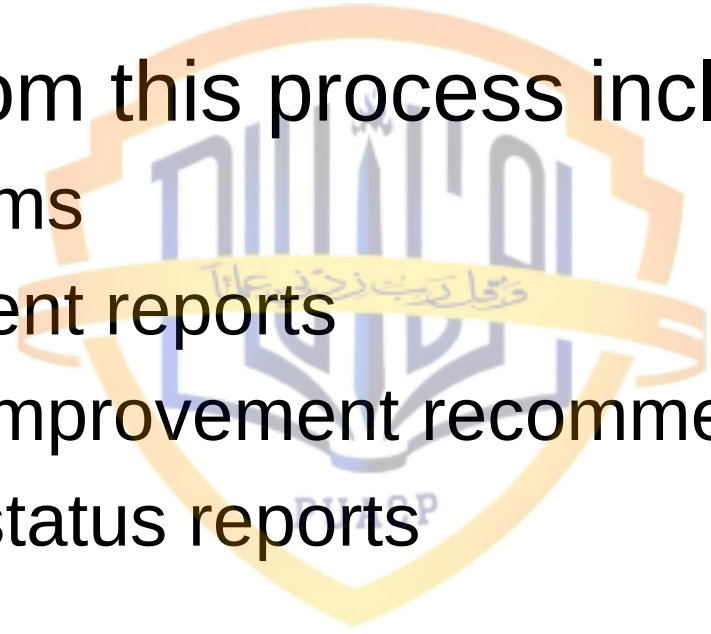
- Inputs include

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- ☐ Assessment checklists
 - ☐ Current processes
 - ☐ ISO 9000
 - ☐ SEI Capability Maturity Model
 - ☐ Information Technology Business Group (ITBG) Testing Capability Maturity Model
 - ☐ Software Process Improvement Capability Determination (SPICE) model

- This has eight (8) sub-processes

Sub-Processes

- Review project plans
- Identify process improvement opportunities
- Develop process improvement plan
- Prepare for assessment
- Perform assessment
- Process assessment result
- Monitor action plan progress
- Update action plans

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- Outputs from this process include
 - Action items
 - Assessment reports
 - Process improvement recommendations
 - Periodic status reports

8. Monitor Test Program

- The final sub-process within the QA process is monitor test program
- This sub-process is designed around an SQA group that does not have responsibility for testing
- It is logical for the SQA organization to take responsibility for assuring that the testing process is performed according to the documented and approved test documentation (test plans, test designs, test cases, and test procedures)

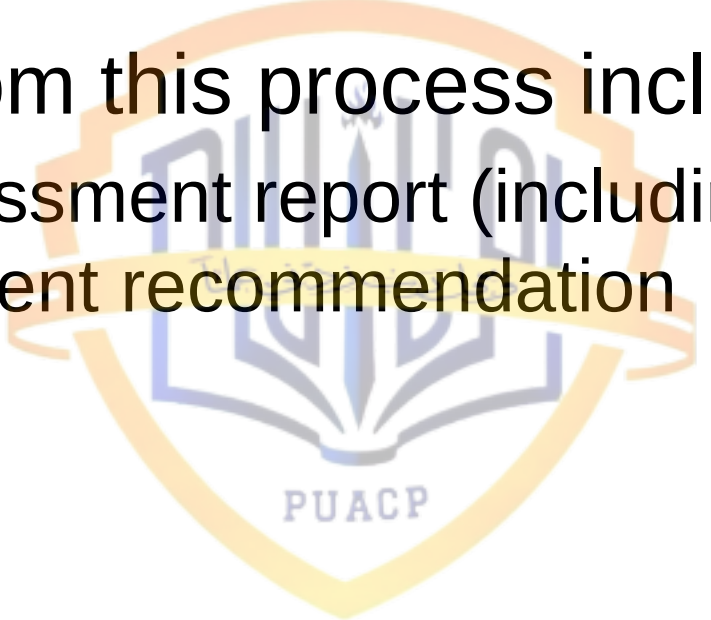
- Inputs include

- ☐ Standards
- ☐ Test documentation
 - Test plan
 - Test design
 - Test cases
 - Test procedures (if applicable)
 - Test summary report
- ☐ Test output data
- ☐ Test process

- There are eight (8) sub-processes

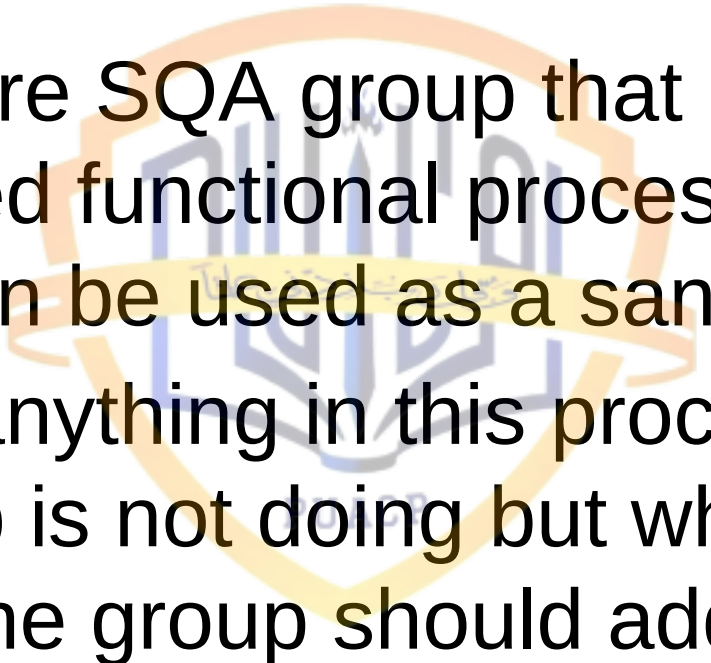
Sub-Processes

- Establish test metrics database
- Collect test metrics
- Report test metrics
- Review test documentation
- Monitor test execution
- Plan test process improvement
- Assess test process
- Develop test assessment report

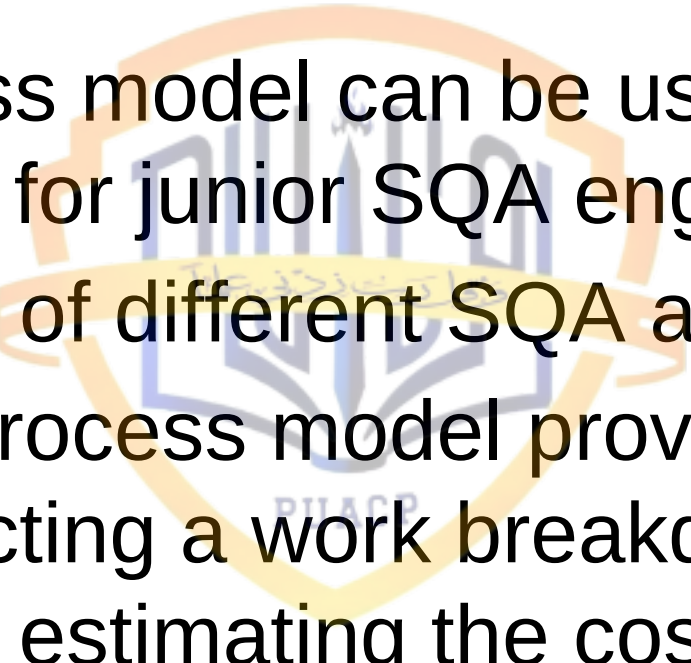
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- Outputs from this process include
 - Test assessment report (including process improvement recommendation)
 - Issues

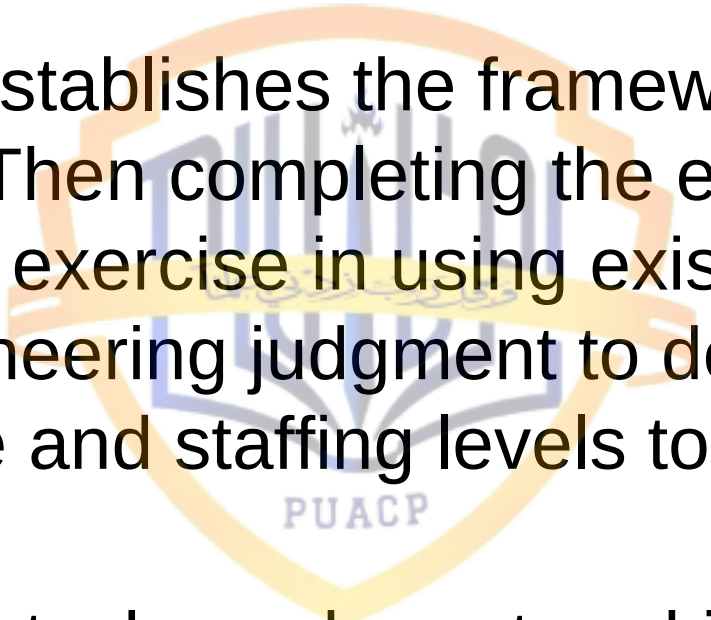
Summary Starts

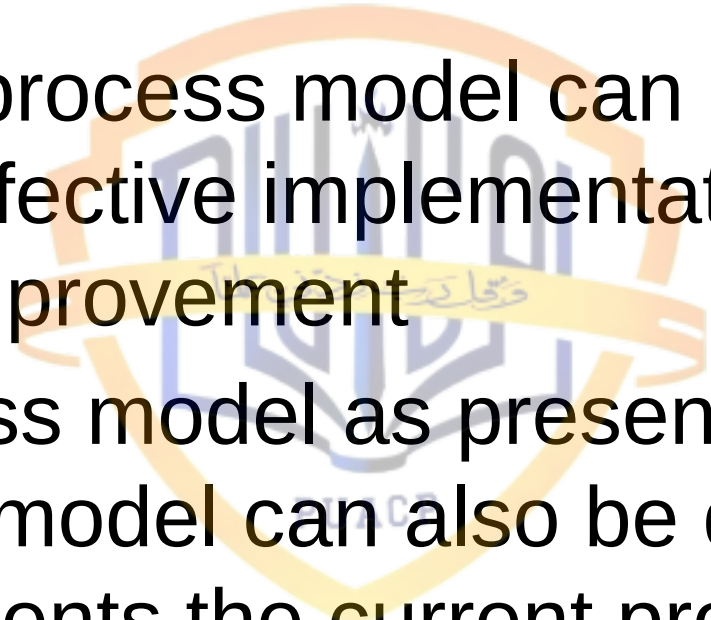
- This top-level model of the SQA/SQE process provides a generic model
- A specific SQA organization should tailor this model to fit its own environment and organization
- Use of this SQA process model depends on the maturity of a specific SQA group

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- For a mature SQA group that has a fully documented functional process, this process can be used as a sanity check
 - If there is anything in this process that the SQA group is not doing but which seems valuable, the group should add that process to its operations

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- For a mature group with a functional process that is not documented, these diagrams could be tailored to match the existing process. Then the process could be formally documented

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- This process model can be used as a training aid for junior SQA engineers
 - Scheduling of different SQA activities
 - The SQA process model provides a tool for constructing a work breakdown structure in estimating the cost and schedule for SQA activities

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- The model establishes the framework and list of SQA tasks. Then completing the estimate becomes an exercise in using existing metrics data or engineering judgment to determine the elapsed time and staffing levels to complete each task
 - Finally, each task can be entered into a schedule tracking package so activities can be scheduled and tracked on an ongoing basis

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- The SQA process model can be used as a guide to effective implementation of process improvement
 - The process model as presented is “ideal”; a process model can also be developed that represents the current process in use by the SQA organization

Summary

- Today, we've discussed a process model for software quality assurance/quality engineering



References

- A process model of Software Quality Assurance/Software Quality Engineering by Roger Drabick, in SQP Vol. 2, No. 4, 2000, ASQ